SYNTHETIC SAMPLE DATA: NOT AN OFFICIAL RECORD. Created for FDE AI Hackathon 5.0.

CS 2100: Program Design and Implementation 1

Institution: Northeastern University **Credits:** 5 (semester hours) **Instructor:** Staff
Prerequisite: CS 2000 **Contact hours:** 4 lecture hours + 1 lab per week, 14 weeks (5 credits including lab CS 2101) **Language:** A teaching language with test-first tooling

Course description

Systematic program design using design recipes, data definitions and test-first development. Emphasises functional decomposition and abstraction; lists and trees are introduced as data, but asymptotic analysis and hash tables are not covered.

Learning outcomes

- Follow a design recipe from data definition to tested function
- Design functions over lists, trees and other self-referential data
- Use abstraction and higher-order functions to remove duplication
- Write unit tests before implementation

Weekly schedule

Week	Topic
1	The design recipe; atomic data
2	Structured data and data definitions
3	Itemizations and unions
4	Self-referential data: lists
5	Functions over lists; testing
6	Trees as data
7	Mutual recursion
8	Midterm; abstraction
9	Higher-order functions (map, filter, fold)
10	Accumulators
11	Generative recursion
12	Introduction to classes and objects
13	Larger program design project
14	Project presentations; final exam

Assessment

Weekly problem sets 40%, labs 10%, midterm 20%, final 30%. Note: asymptotic analysis and hash tables are covered in CS 2700.

Textbook: instructor-provided open course notes.

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